In SEM, there are several electromagnetic lenses, including condenser lenses and one objective lens. Electromagnetic lenses are for electron probe formation, not for image formation directly, as in TEM. Two condenser lenses reduce the crossover diameter of the electron beam. The objective lens further reduces the cross-section of the electron beam and focuses the electron beam as probe on the specimen surface (Figure 2.10). Objective lens thus functions like a condenser. This is in contrast to TEM where objective lens does the magnification. Major difference between SEM and TEM are given in Table 2.2. SEMs are equipped with an energy dispersive spectrometer (EDS) detection system which is able to detect and display most of the X-ray spectrum. The detector normally consists of semiconducting silicon or germanium.

Scanning transmission electron microscopy (STEM) combines the principles of transmission electron microscopy and scanning electron microscopy and can be performed on either type of instrument. Like TEM, STEM requires very thin samples and looks primary at beam electrons transmitted by the sample. One of its principal advantages over TEM is in enabling the use of other of signals that cannot be spatially correlated in TEM, including secondary electrons, scattered beam electrons, characteristic X-rays, and electron energy loss.

Summary

Microscopes are specialized optical instruments designed to produce magnified visual or photographic images of objects or specimens that are too small to be seen with naked eye. Frits Zernike a Dutch Physicist invented the Phase Contrast Microscope and was awarded Nobel Prize in 1953. It is the first microscopic method which allows the observation of living cell. The image of the aperture is formed at the rear focal plane of the objective. In this plane there is a phase shifting element or phase plate. Deviated rays from object form structures due to different refractive index. Light waves that are in phase (that is, their peaks and valleys exactly coincide) reinforce one another and their total intensity increases. Light waves that are out of phase by exactly one-half wavelength cancel each other and result in no intensity. Fluorescence microscopy is a very powerful analytical tool that combines the magnifying properties of light microscopy with visualization of fluorescence. Examining the ultra structure of cellular components such as nucleus, plasma membrane, mitochondria and others requires 10,000x plus magnification which was just not possible using Light Microscopes. It is first built by Knoll in 1935. It is used to study the three dimensional images of the surfaces of cells, tissues or particles. The SEM allows viewing the surfaces of specimens without sectioning. The specimen is first fixed in liquid propane at-180°C and then dehydrated in alcohol at-70°C. **Scanning transmission electron microscopy (STEM)** combines the principles of transmission electron microscopy and scanning electron microscopy and can be performed on either type of instrument. One of its principal advantages over TEM

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